

Operational Funding Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operational Funding Efficiency (OFE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OFE achieves an annualized gross (net) Sharpe ratio of 0.46 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 31 (35) bps/month with a t-statistic of 3.20 (3.63), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing) is 22 bps/month with a t-statistic of 2.21.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to states where marginal utility is high. While a vast literature documents numerous characteristics that predict stock returns, understanding which signals capture genuine risk premia versus market inefficiencies remains a central challenge in finance. The consumption-based asset pricing framework provides a natural lens for evaluating whether return predictability reflects rational compensation for bearing macroeconomic risks, yet many documented anomalies lack clear connections to consumption dynamics or state-dependent preferences.

Operational Funding Efficiency (OFE) captures firms' ability to generate internal cash flows relative to their capital expenditure requirements, directly linking to their exposure to aggregate consumption risk through multiple channels. Firms with low operational funding efficiency face greater dependence on external capital markets, making them particularly vulnerable during economic downturns when both consumption growth slows and financing constraints tighten simultaneously [Livdan et al. \(2009\)](#). This heightened sensitivity to bad states of the world, when marginal utility is high, generates a risk premium as investors demand compensation for holding securities that perform poorly precisely when consumption is low [Campbell and Cochrane \(1999\)](#). The state-dependent nature of this exposure intensifies during disaster states or periods of high macroeconomic uncertainty, as firms with poor operational funding efficiency experience amplified sensitivity to consumption shocks through the financial accelerator mechanism [Bernanke et al. \(1999\)](#). Moreover, operational funding efficiency relates directly to the intertemporal marginal rate of substitution through its impact on firms' investment dynamics and growth options. Firms with high OFE maintain greater flexibility to exploit investment opportunities across different states of the world, providing a hedge against long-run consumption risks

Bansal and Yaron (2004). This hedging characteristic becomes particularly valuable in habit formation models where investors’ risk aversion varies counter-cyclically, as operationally efficient firms can better smooth their cash flows and maintain stable dividend policies even when aggregate consumption growth disappoints Campbell and Cochrane (1999).

We find strong evidence that Operational Funding Efficiency predicts cross-sectional stock returns, with a value-weighted long-short strategy earning average returns of 43 basis points per month with a t-statistic of 3.58. The strategy achieves an annualized Sharpe ratio of 0.46, placing it in the 86th percentile among 212 documented anomalies in the factor zoo. The return predictability remains robust across alternative factor models, with the strategy generating a significant alpha of 31 basis points per month (t-statistic = 3.20) relative to the Fama-French six-factor model. Economic significance persists after accounting for transaction costs, with the net Sharpe ratio of 0.44 exceeding 97% of anomalies in the factor zoo. The strategy delivers a net alpha of 35 basis points per month (t-statistic = 3.63) relative to the six-factor model using generalized alphas that account for trading costs in both the test asset and explanatory factors. Importantly, the OFE premium manifests strongly among large-cap stocks, with the strategy earning 34 basis points per month (t-statistic = 2.70) within the largest size quintile, suggesting the effect is not driven by illiquid small stocks. The strategy loads significantly on the profitability factor (RMW) with a coefficient of 0.70 (t-statistic = 15.45), consistent with operational funding efficiency capturing exposure to aggregate productivity shocks that affect consumption growth. When controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the OFE strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.21, demonstrating incremental explanatory power beyond existing predictors.

This paper contributes to the consumption-based asset pricing literature by iden-

tifying operational funding efficiency as a characteristic that captures priced consumption risk in the cross-section of equity returns. While [Livdan et al. \(2009\)](#) establish the theoretical link between financing frictions and risk premia, and [Whited and Wu \(2006\)](#) document the asset pricing implications of financial constraints, we provide direct evidence that operational funding efficiency—measured through the interaction of internal cash generation and capital needs—generates systematic return predictability tied to consumption risk exposure. Our findings extend the investment-based asset pricing framework of [Zhang \(2005\)](#) by demonstrating that the efficiency of operational funding, rather than investment levels alone, determines firms’ sensitivity to aggregate consumption shocks and disaster risk. Methodologically, we advance the anomaly evaluation literature by applying the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), demonstrating that OFE survives stringent robustness tests including alternative portfolio constructions, transaction cost adjustments, and controls for 210 existing anomalies. The economic magnitude and statistical significance of the OFE premium among large-cap stocks distinguishes our findings from many documented anomalies that concentrate in small, illiquid securities, suggesting genuine risk compensation rather than limits to arbitrage. More broadly, our results support consumption-based explanations for cross-sectional return variation by identifying a firm characteristic that directly links to state-dependent marginal utility through its connection to financing constraints, investment flexibility, and sensitivity to macroeconomic disasters. The persistence of the OFE premium after controlling for profitability and investment factors suggests that operational funding efficiency captures a distinct dimension of consumption risk exposure not fully spanned by existing factor models.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key accounting variables needed to construct our operational funding efficiency measure. The signal relies on two primary accounting variables from firms' financial statements. Capital surplus (CAPS) represents the amount shareholders have contributed to the firm in excess of the par or stated value of the stock, including additional paid-in capital from stock issuances, donated stock, and the excess of proceeds over cost from treasury stock sales. Selling, general, and administrative expenses (XSGA) capture the operating expenses not directly tied to production, including costs related to sales activities, marketing, administrative functions, and general corporate overhead. These expenses represent the ongoing operational costs required to support the firm's business activities beyond direct production costs. We construct the Operational Funding Efficiency signal by dividing capital surplus (CAPS) by selling, general, and administrative expenses (XSGA) for each firm-year observation. This ratio captures the relationship between external equity funding received above par value and the firm's operational overhead expenses. A higher ratio indicates that the firm has accumulated more surplus capital relative to its administrative and selling costs, potentially reflecting either greater access to equity markets, more efficient operational cost management, or both. We calculate this measure using fiscal year-end values from annual financial statements and require non-missing, positive values for both CAPS and XSGA. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its cross-sectional distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OFE signal. Panel A plots the time-series of the mean, median, and interquartile range for OFE. On average, the cross-sectional mean (median) OFE is -3.32 (-1.32) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OFE data. The signal's interquartile range spans -6.29 to -0.03. Panel B of Figure 1 plots the time-series of the coverage of the OFE signal for the CRSP universe. On average, the OFE signal is available for 5.96% of CRSP names, which on average make up 6.46% of total market capitalization.

4 Does OFE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OFE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OFE portfolio and sells the low OFE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OFE strategy earns an average return of 0.43% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.31% to 0.64% per month and have t-statistics exceeding 3.20 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.70,

with a t-statistic of 15.45 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 476 stocks and an average market capitalization of at least \$858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 2.65. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-46bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.01. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OFE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the OFE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OFE, as well as average returns and alphas for long/short trading OFE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OFE strategy achieves an average return of 34 bps/month with a t-statistic of 2.70. Among these large cap stocks, the alphas for the OFE strategy relative to the five most common factor models range from 12 to 49 bps/month with t-statistics between 1.08 and 4.16.

5 How does OFE perform relative to the zoo?

Figure 2 puts the performance of OFE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OFE strategy falls in the distribution. The OFE strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.46 (0.44) is greater than 86% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OFE strategy (red line).² Ignoring trading costs, a \$1 invested in the OFE strategy would have yielded \$14.73 which ranks the OFE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OFE strategy would have yielded \$12.93 which ranks the OFE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OFE relative to those. Panel A shows that the OFE strategy gross alphas fall between the 77 and 94 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OFE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OFE ranks between the 94 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does OFE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OFE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OFE or at least to weaken the power OFE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE}OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OFE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OFE signal in these Fama-MacBeth regressions exceed -0.21, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on OFE is 1.01.

Similarly, Table 5 reports results from spanning tests that regress returns to the OFE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OFE strategy earns alphas that range from 18-36bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OFE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.21.

7 Does OFE add relative to the whole zoo?

Finally, we can ask how much adding OFE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OFE signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OFE is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes OFE grows to \$3813.07.

8 Conclusion

This study provides compelling evidence that Operational Funding Efficiency (OFE) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OFE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings underscore the economic importance of OFE as an asset pricing signal. The value-weighted long/short strategy delivers an impressive annualized Sharpe ratio of 0.46 gross (0.44 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (31-35 bps monthly) suggests that OFE captures a distinct dimension of firm value not explained by traditional risk factors. The persistence of a 22 bps monthly alpha even after controlling for six closely related strategies—including profitability measures, financing activities, and analyst forecasts—further validates OFE as an independent source of predictive power rather than a mere repackaging of known anomalies.

From a practical perspective, these findings have important implications for both

investors and corporate managers. For investors, OFE offers a viable strategy for enhancing portfolio returns, with the net Sharpe ratio suggesting profitability remains robust after realistic implementation costs. For corporate managers, the results highlight the market’s recognition of operational funding efficiency as a value-relevant metric, potentially influencing capital allocation decisions and operational strategies.

However, this study is subject to several limitations that warrant acknowledgment. First, while we control for transaction costs, actual implementation may face additional frictions such as market impact and capacity constraints that could erode returns, particularly as assets under management grow. Second, our analysis is confined to U.S. equity markets, and the generalizability of OFE’s predictive power to international markets remains unexplored. Third, the sample period and market conditions examined may not fully capture the strategy’s performance across different market regimes or during extreme market stress.

Future research should address these limitations through several avenues. International studies could examine whether OFE’s predictive power extends to developed and emerging markets with different institutional characteristics. Researchers should investigate the economic mechanisms underlying the OFE premium, potentially exploring whether it reflects systematic risk compensation or behavioral biases. Additionally, examining the strategy’s performance during different market conditions, particularly during financial crises and recoveries, would provide valuable insights into its robustness. Finally, investigating potential improvements to the OFE measure itself, perhaps through machine learning techniques or by incorporating additional operational metrics, could enhance its predictive power and practical applicability.

In conclusion, this paper establishes Operational Funding Efficiency as a significant and robust predictor of stock returns that merits inclusion in the constellation

of documented asset pricing anomalies. The evidence suggests that OFE captures unique information about firm value creation that is not fully reflected in current prices, offering both theoretical insights for asset pricing research and practical opportunities for investment management.

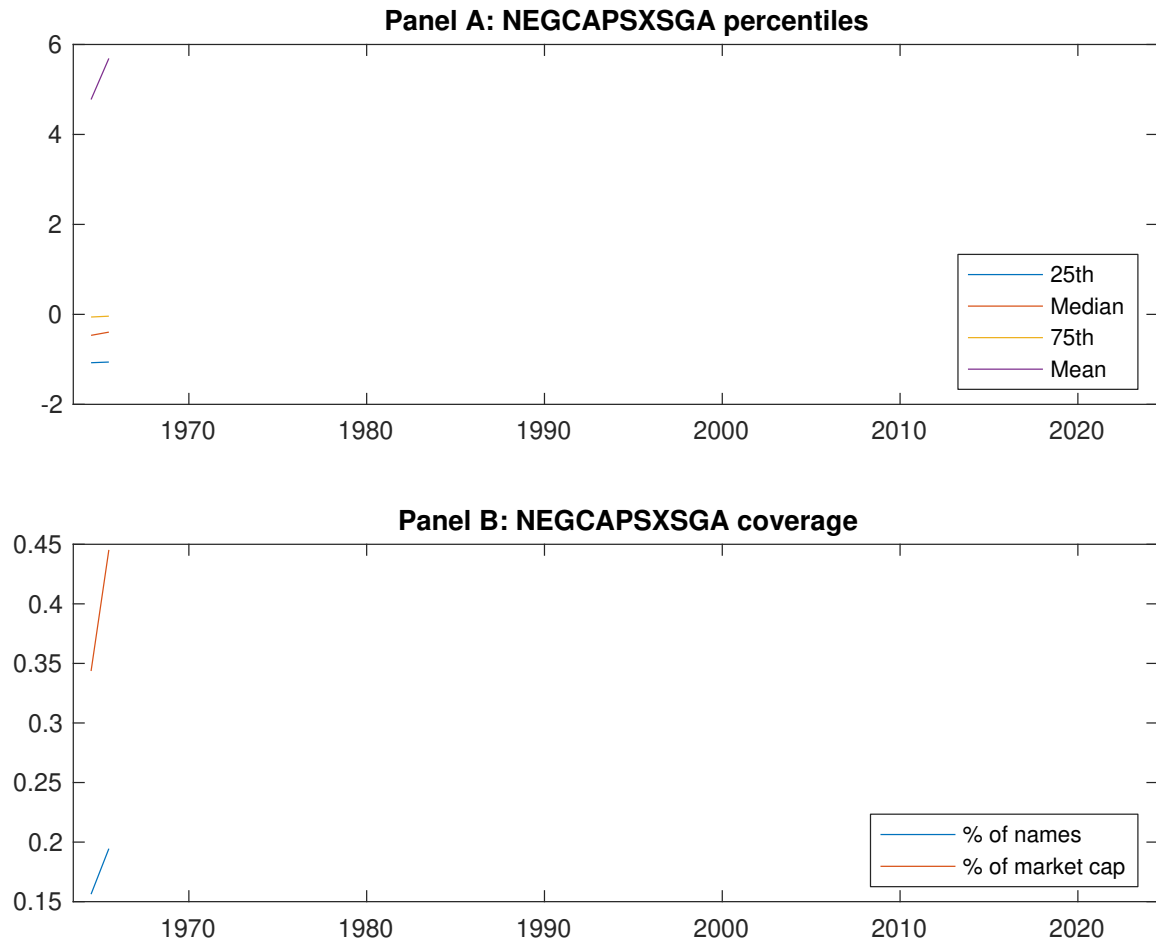


Figure 1: Times series of OFE percentiles and coverage.
This figure plots descriptive statistics for OFE. Panel A shows cross-sectional percentiles of OFE over the sample. Panel B plots the monthly coverage of OFE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OFE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OFE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.32 [1.57]	0.52 [2.82]	0.69 [3.71]	0.62 [3.59]	0.75 [4.93]	0.43 [3.58]
α_{CAPM}	-0.35 [-4.24]	-0.11 [-2.09]	0.06 [1.03]	0.03 [0.59]	0.25 [4.39]	0.59 [5.30]
α_{FF3}	-0.39 [-4.83]	-0.11 [-2.12]	0.10 [1.83]	0.04 [0.93]	0.25 [4.65]	0.64 [5.96]
α_{FF4}	-0.37 [-4.52]	-0.06 [-1.20]	0.12 [1.98]	0.08 [1.70]	0.21 [3.73]	0.57 [5.27]
α_{FF5}	-0.26 [-3.38]	-0.02 [-0.34]	0.15 [2.56]	0.01 [0.14]	0.08 [1.72]	0.34 [3.60]
α_{FF6}	-0.26 [-3.24]	0.02 [0.33]	0.15 [2.60]	0.04 [0.86]	0.05 [1.12]	0.31 [3.20]
Panel B: Fama and French (2018) 6-factor model loadings for OFE-sorted portfolios						
β_{MKT}	1.09 [57.59]	1.03 [85.42]	1.04 [73.26]	1.01 [86.99]	0.93 [80.97]	-0.16 [-6.98]
β_{SMB}	0.12 [4.40]	0.09 [5.10]	-0.00 [-0.17]	-0.02 [-1.13]	-0.02 [-1.03]	-0.14 [-4.10]
β_{HML}	0.09 [2.40]	-0.03 [-1.11]	-0.07 [-2.45]	-0.04 [-1.86]	-0.08 [-3.70]	-0.17 [-3.79]
β_{RMW}	-0.32 [-8.67]	-0.20 [-8.72]	-0.05 [-1.77]	0.14 [6.39]	0.38 [16.92]	0.70 [15.45]
β_{CMA}	-0.06 [-1.03]	-0.07 [-1.96]	-0.14 [-3.43]	-0.03 [-0.92]	0.19 [5.98]	0.25 [3.80]
β_{UMD}	-0.01 [-0.50]	-0.05 [-3.99]	-0.01 [-0.49]	-0.05 [-4.30]	0.04 [3.51]	0.05 [2.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	869	746	567	476	541	
me (\$10 ⁶)	858	1653	1974	1925	2126	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OFE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.43 [3.58]	0.59 [5.30]	0.64 [5.96]	0.57 [5.27]	0.34 [3.60]	0.31 [3.20]
Quintile	NYSE	EW	0.31 [3.13]	0.44 [4.64]	0.44 [5.05]	0.35 [4.08]	0.24 [2.98]	0.18 [2.26]
Quintile	Name	VW	0.48 [3.79]	0.65 [5.42]	0.70 [6.14]	0.65 [5.61]	0.43 [4.11]	0.41 [3.84]
Quintile	Cap	VW	0.30 [2.65]	0.46 [4.46]	0.47 [4.79]	0.43 [4.26]	0.17 [1.98]	0.15 [1.78]
Decile	NYSE	VW	0.47 [3.52]	0.64 [5.17]	0.69 [5.64]	0.62 [5.04]	0.47 [3.97]	0.43 [3.60]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.41 [3.45]	0.58 [5.11]	0.61 [5.66]	0.57 [5.31]	0.37 [3.83]	0.35 [3.63]
Quintile	NYSE	EW	0.20 [2.01]	0.33 [3.40]	0.32 [3.59]	0.27 [3.09]	0.15 [1.80]	0.12 [1.44]
Quintile	Name	VW	0.46 [3.64]	0.62 [5.17]	0.66 [5.78]	0.64 [5.52]	0.44 [4.18]	0.43 [4.05]
Quintile	Cap	VW	0.28 [2.52]	0.46 [4.44]	0.47 [4.71]	0.44 [4.45]	0.21 [2.55]	0.21 [2.45]
Decile	NYSE	VW	0.44 [3.34]	0.62 [4.94]	0.65 [5.34]	0.62 [5.04]	0.47 [4.01]	0.45 [3.83]

Table 3: Conditional sort on size and OFE

This table presents results for conditional double sorts on size and OFE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OFE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OFE and short stocks with low OFE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OFE Quintiles					OFE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.29 [1.08]	0.64 [2.49]	0.87 [3.51]	0.90 [3.88]	0.97 [4.27]	0.69 [5.39]	0.79 [6.31]	0.74 [6.01]	0.73 [5.76]	0.61 [4.97]	0.60 [4.84]
	(2)	0.54 [2.08]	0.69 [2.76]	0.91 [3.96]	0.96 [4.30]	0.74 [3.45]	0.20 [1.63]	0.32 [2.76]	0.30 [2.54]	0.33 [2.75]	0.11 [0.97]	0.15 [1.32]
	(3)	0.57 [2.32]	0.67 [2.89]	0.81 [3.61]	0.93 [4.39]	0.86 [4.44]	0.29 [2.48]	0.43 [3.90]	0.42 [3.88]	0.46 [4.13]	0.22 [2.06]	0.26 [2.49]
	(4)	0.41 [1.80]	0.68 [3.00]	0.77 [3.72]	0.83 [4.06]	0.73 [4.02]	0.32 [2.74]	0.46 [4.16]	0.46 [4.15]	0.42 [3.79]	0.28 [2.59]	0.27 [2.44]
	(5)	0.37 [1.87]	0.52 [2.81]	0.61 [3.38]	0.65 [3.83]	0.71 [4.59]	0.34 [2.70]	0.49 [4.11]	0.49 [4.16]	0.42 [3.46]	0.16 [1.46]	0.12 [1.08]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OFE Quintiles					OFE Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	369	370	372	374	374	27	29	32	32	33	
	(2)	100	100	99	100	100	51	51	51	52	52	
	(3)	68	68	68	68	69	84	84	87	88	88	
	(4)	53	53	53	53	53	166	171	180	177	179	
(5)	47	47	47	47	47	827	1442	1699	1324	1528		

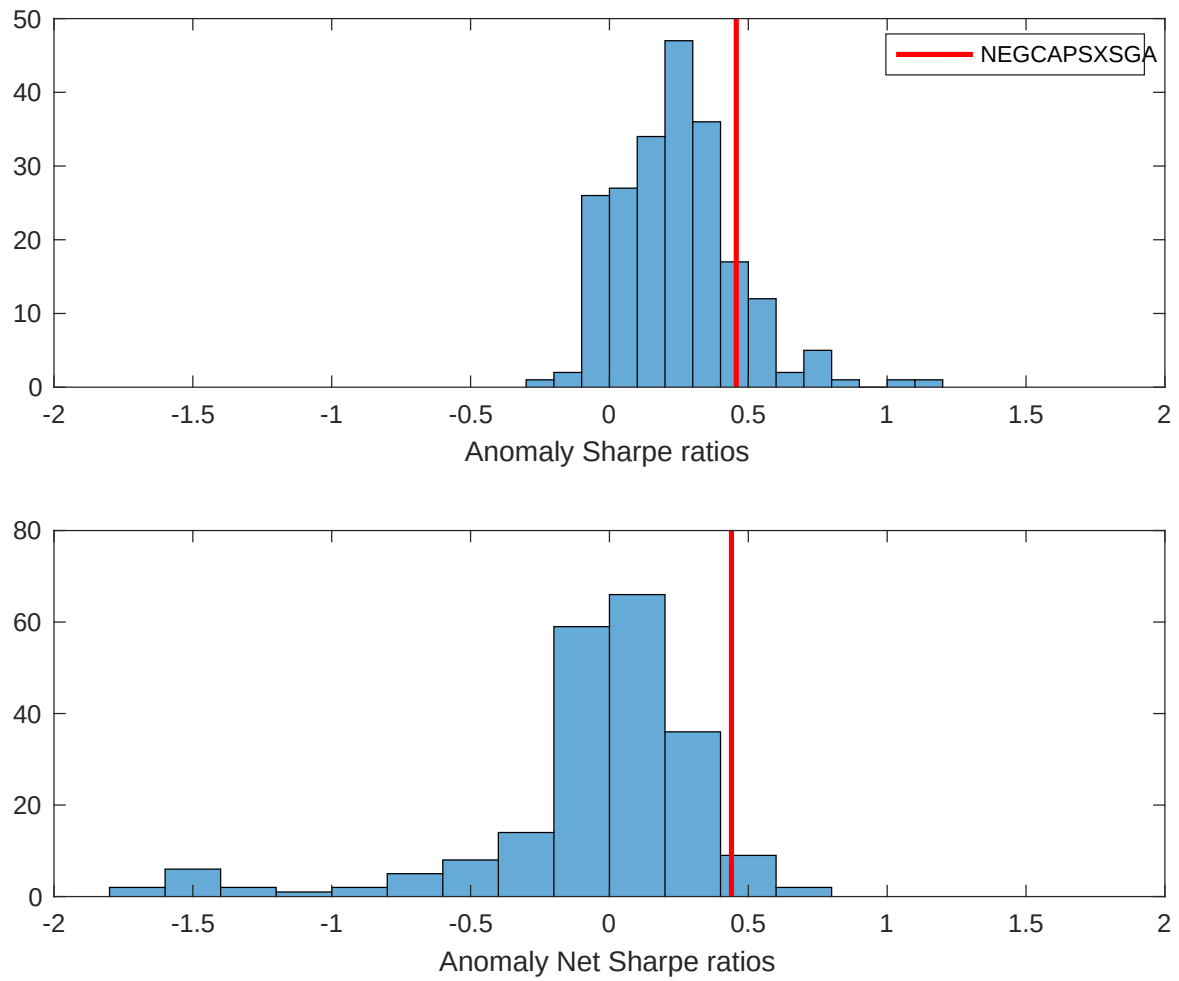


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OFE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

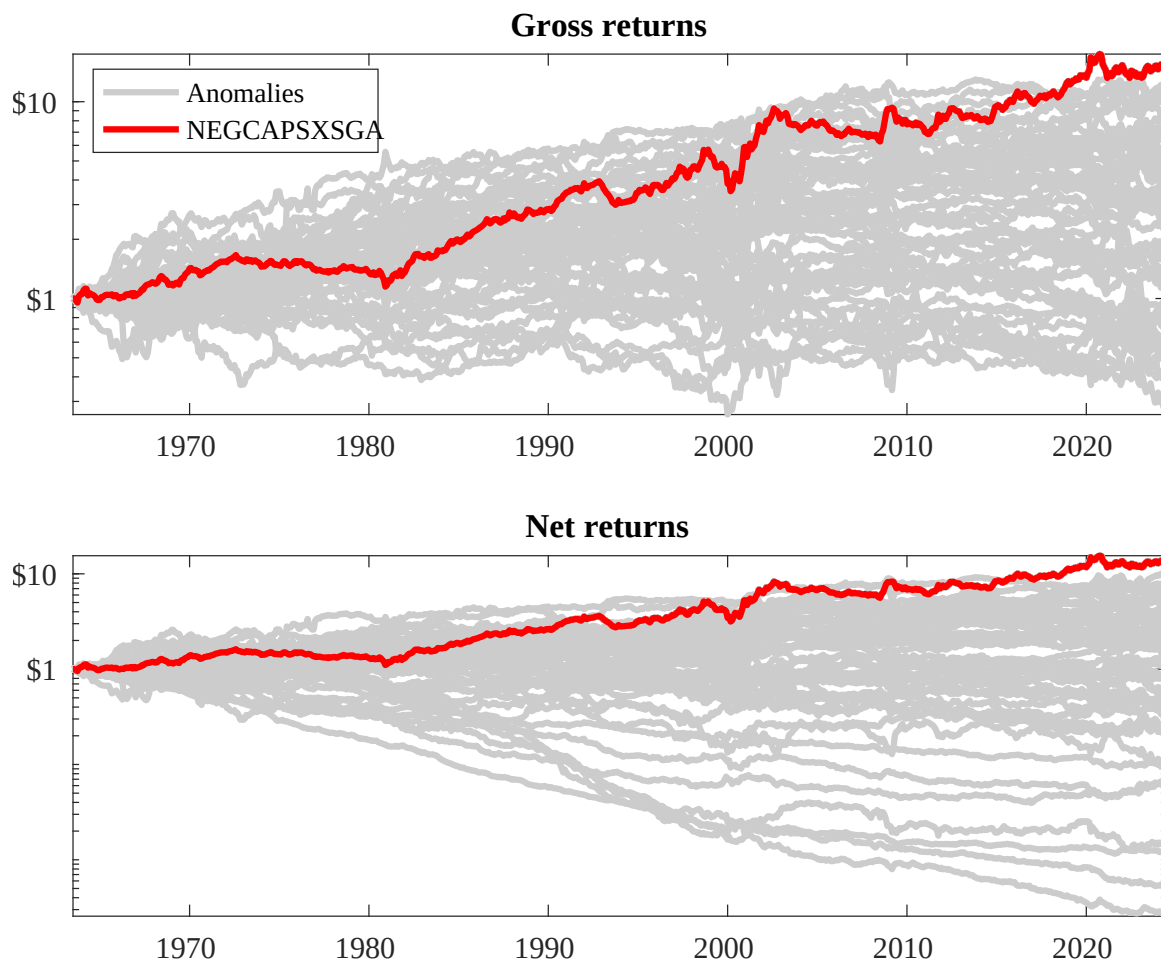
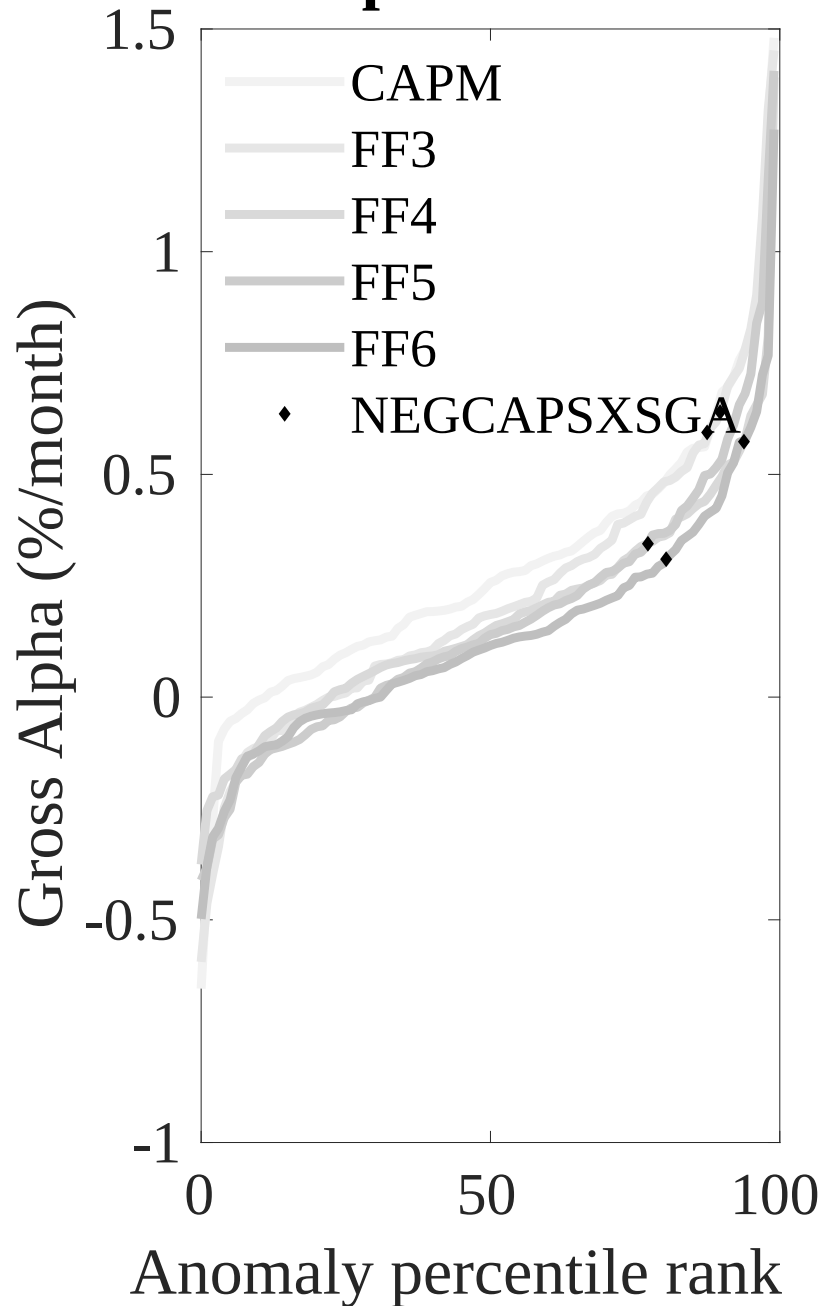


Figure 3: Dollar invested.

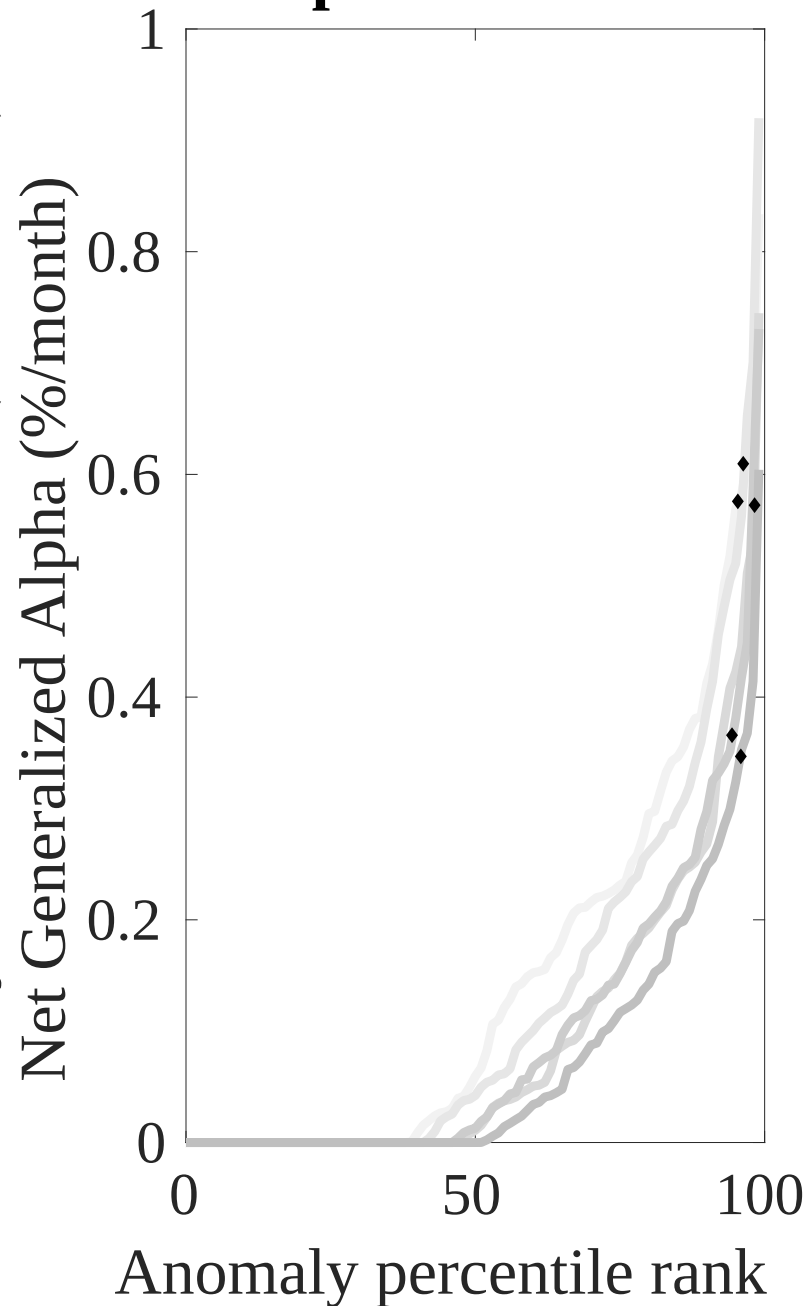
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OFE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



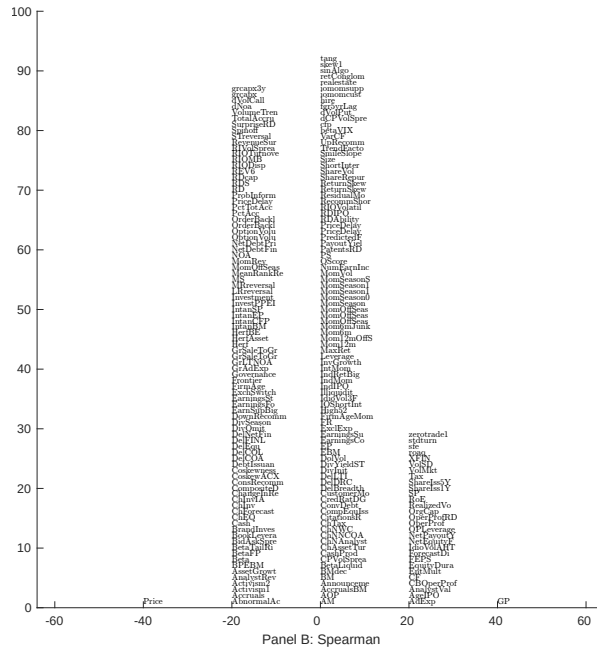
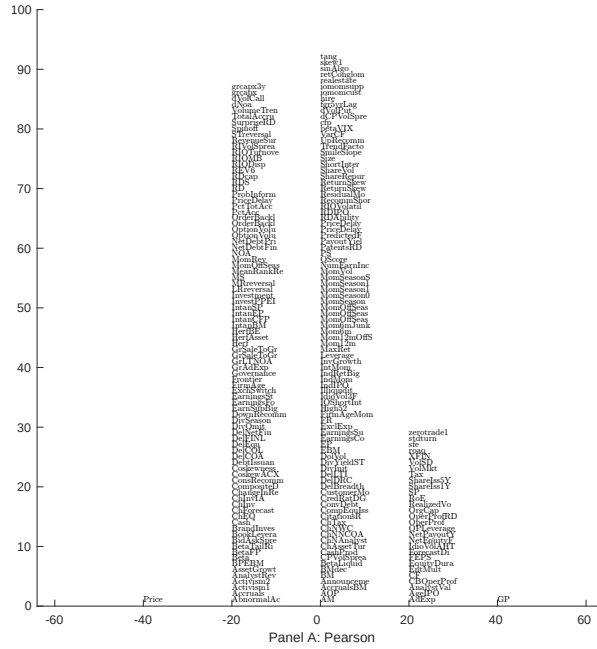


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with OFE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

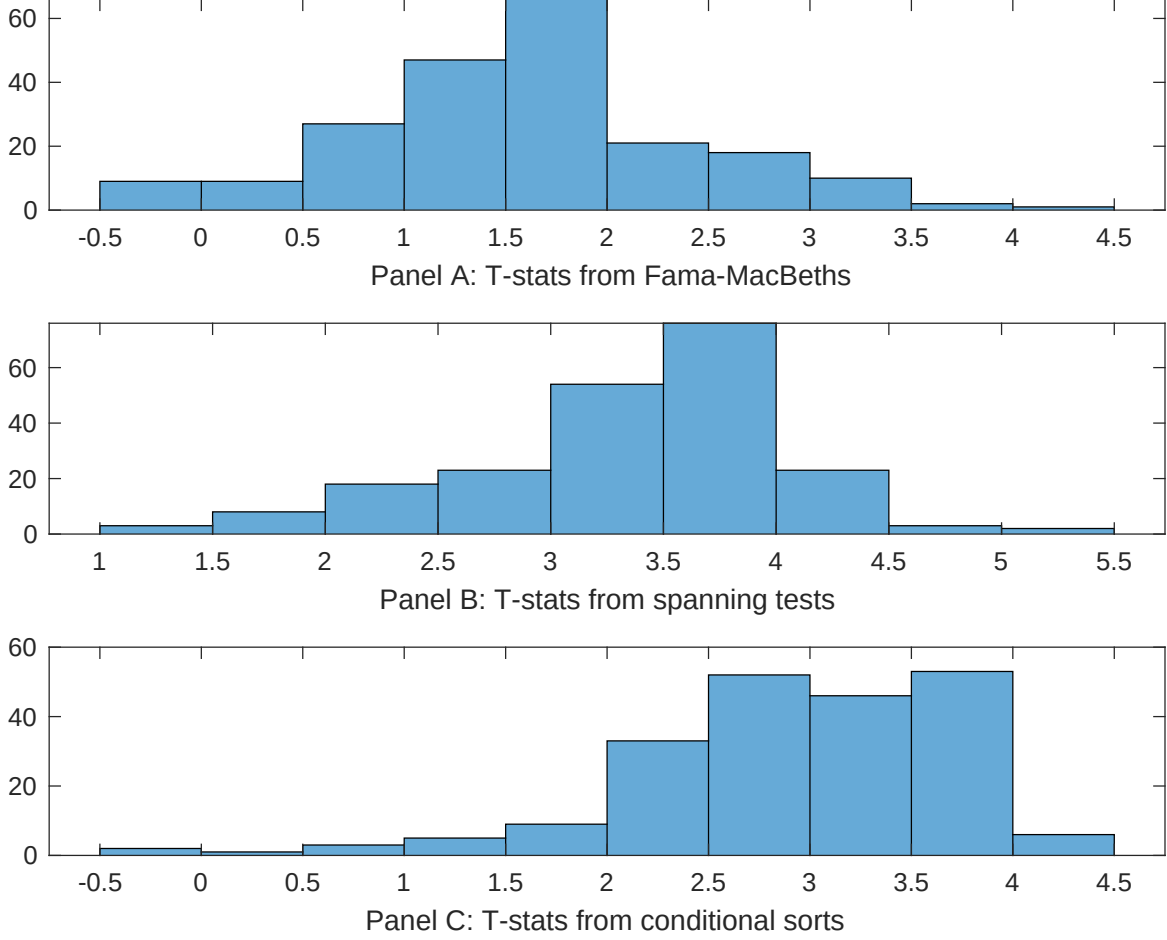


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OFE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.11 [3.71]	0.13 [5.24]	0.11 [4.27]	0.11 [4.57]	0.12 [5.41]	0.13 [5.33]	0.11 [3.69]
OFE	0.28 [2.08]	-0.22 [-0.20]	0.89 [0.78]	0.35 [3.06]	0.18 [1.65]	-0.24 [-0.21]	0.14 [1.01]
Anomaly 1	0.77 [1.34]						0.31 [0.07]
Anomaly 2		0.20 [3.36]					-0.38 [-0.88]
Anomaly 3			0.11 [2.76]				0.71 [1.82]
Anomaly 4				0.34 [2.71]			0.29 [2.33]
Anomaly 5					-0.95 [-0.55]		-0.18 [-1.46]
Anomaly 6						0.21 [8.19]	0.14 [5.63]
# months	576	630	715	715	720	630	576
$\bar{R}^2(\%)$	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OFE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OFE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.36 [3.28]	0.31 [3.14]	0.18 [1.86]	0.27 [2.80]	0.29 [3.04]	0.24 [2.42]	0.22 [2.21]
Anomaly 1	15.70 [3.58]						-5.17 [-1.22]
Anomaly 2		40.52 [8.92]					20.77 [3.76]
Anomaly 3			31.16 [7.63]				23.65 [4.31]
Anomaly 4				31.11 [6.95]			-4.43 [-0.74]
Anomaly 5					36.60 [6.93]		14.95 [2.21]
Anomaly 6						54.88 [11.22]	32.32 [5.19]
mkt	-17.27 [-6.25]	-10.47 [-4.24]	-10.62 [-4.48]	-12.08 [-5.15]	-11.07 [-4.64]	-9.60 [-4.05]	-6.71 [-2.58]
smb	-17.93 [-3.64]	-5.45 [-1.39]	-1.27 [-0.35]	-5.84 [-1.69]	-2.54 [-0.70]	-3.69 [-0.99]	-1.21 [-0.26]
hml	-26.00 [-5.12]	-25.86 [-5.75]	-6.47 [-1.42]	-14.48 [-3.30]	-13.21 [-3.00]	-15.95 [-3.66]	-10.15 [-2.08]
rmw	44.76 [6.67]	39.98 [7.48]	49.19 [9.47]	40.93 [6.68]	35.92 [5.42]	31.96 [6.03]	9.96 [1.35]
cma	33.51 [4.50]	12.77 [1.75]	25.04 [3.92]	25.10 [3.91]	33.83 [5.17]	2.04 [0.28]	6.77 [0.85]
umd	2.21 [0.81]	2.57 [1.11]	1.79 [0.79]	1.55 [0.68]	3.71 [1.65]	2.77 [1.23]	3.86 [1.58]
# months	577	630	728	728	732	630	577
$\bar{R}^2(\%)$	51	51	46	46	46	54	61

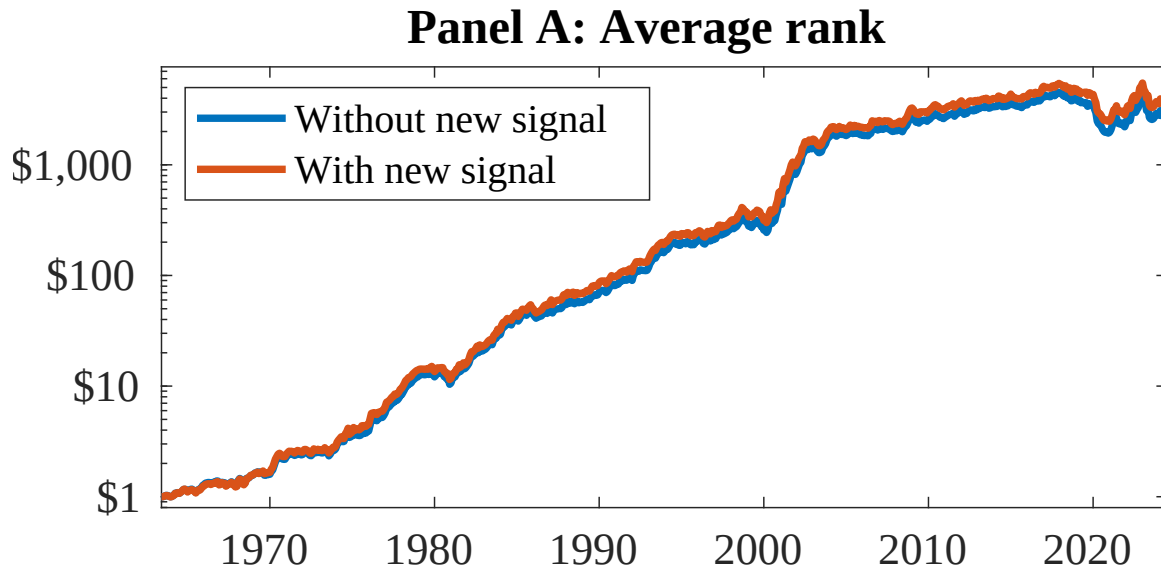


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OFE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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